Ultra-Simplified Kinematic Equations

Auto-Generated with TrigSimp

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Note on Simplification

These equations have been processed using aggressive trigonometric simplification (Fu's method). Identities like $\sin^2 x + \cos^2 x = 1$ and $\cos(a+b) = \cos a \cos b - \sin a \sin b$ have been applied automatically.

1 Permutation 1: $T_1T_2T_3T_4 = T_7^{-1}T_6^{-1}T_5^{-1}$

$$
\begin{aligned}
(\text{n}\dot{}\text{x}) &: -\sin(\theta_4)\sin(\theta_1+\theta_2) + \cos(\theta_4)\cos(\theta_1+\theta_2)\cos(\theta_3+1.0\pi) \\
&= -\sin(\theta_5)\sin(\theta_7) + \cos(\theta_5)\cos(\theta_7)\cos(\theta_6+1.0\pi) \\
(\text{o}\dot{}\text{x}) &: -\sin(\theta_3+1.0\pi)\cos(\theta_1+\theta_2) \\
&= \sin(\theta_5)\cos(\theta_7)\cos(\theta_6+1.0\pi) + \sin(\theta_7)\cos(\theta_5) \\
(\text{a}\dot{}\text{x}) &: -\sin(\theta_4)\cos(\theta_1+\theta_2)\cos(\theta_3+1.0\pi) - \sin(\theta_1+\theta_2)\cos(\theta_4) \\
&= \sin(\theta_6+1.0\pi)\cos(\theta_7) \\
(\text{p}\dot{}\text{x}) &: L\left(-2.5\sin(\theta_4)\sin(\theta_1+\theta_2) + 1.0\cos(\theta_1) + 2.5\cos(\theta_4)\cos(\theta_1+\theta_2)\cos(\theta_3+1.0\pi) + 2.0\cos(\theta_1+\theta_2)\right) \\
&= L\left(2.5\sin(\theta_7) - 2.5\cos(\theta_7)\cos(\theta_6+1.0\pi) - 2.0\right) \\
(\text{n}\dot{}\text{y}) &: \sin(\theta_4)\cos(\theta_1+\theta_2) + \sin(\theta_1+\theta_2)\cos(\theta_4)\cos(\theta_3+1.0\pi) \\
&= \sin(\theta_6+1.0\pi)\cos(\theta_5) \\
(\text{o}\dot{}\text{y}) &: -\sin(\theta_1+\theta_2)\sin(\theta_3+1.0\pi) \\
&= -\sin(\theta_5)\sin(\theta_6) \\
(\text{a}\dot{}\text{y}) &: -\sin(\theta_4)\sin(\theta_1+\theta_2)\cos(\theta_3+1.0\pi) + \cos(\theta_4)\cos(\theta_1+\theta_2) \\
&= \cos(\theta_6) \\
(\text{p}\dot{}\text{y}) &: L\left(1.0\sin(\theta_1) + 2.5\sin(\theta_4)\cos(\theta_1+\theta_2) + 2.5\sin(\theta_1+\theta_2)\cos(\theta_4)\cos(\theta_3+1.0\pi) + 2.0\sin(\theta_1+\theta_2)\right) \\
&= 2.5L\sin(\theta_6) \\
(\text{n}\dot{}\text{z}) &: \sin(\theta_3)\cos(\theta_4) \\
&= -\sin(\theta_5)\cos(\theta_7) - \sin(\theta_7)\cos(\theta_5)\cos(\theta_6+1.0\pi) \\
(\text{o}\dot{}\text{z}) &: \cos(\theta_3) \\
&= -\sin(\theta_5)\sin(\theta_7)\cos(\theta_6+1.0\pi) + \cos(\theta_5)\cos(\theta_7) \\
(\text{a}\dot{}\text{z}) &: -\sin(\theta_3)\sin(\theta_4) \\
&= \sin(\theta_6)\sin(\theta_7)
\end{aligned}
$$

$$
\begin{aligned}
(\mathrm{p}\dot{}\mathrm{z}) &: L\left(2.5\sin\left(\theta_3\right)\cos\left(\theta_4\right)+5.0\right) \\
&= L\left(2.5\sin\left(\theta_7\right)\cos\left(\theta_6+1.0\pi\right)+2.5\cos\left(\theta_7\right)\right)
\end{aligned}
$$

2 Permutation 2: $T_2T_3T_4T_5 = T_1^{-1}T_7^{-1}T_6^{-1}$

$$
\begin{aligned}
(\text{n}\dot{}\text{x}) &: (-\sin(\theta_2)\sin(\theta_4) - \cos(\theta_2)\cos(\theta_3)\cos(\theta_4))\cos(\theta_5) + \sin(\theta_3)\sin(\theta_5)\cos(\theta_2) \\
&= \sin(\theta_1)\sin(\theta_6 + 1.0\pi) + \cos(\theta_1)\cos(\theta_7)\cos(\theta_6 + 1.0\pi) \\
(\text{o}\dot{}\text{x}) &: -\sin(\theta_2)\cos(\theta_4) + \sin(\theta_4)\cos(\theta_2)\cos(\theta_3) \\
&= -\sin(\theta_1)\cos(\theta_6 + 1.0\pi) + \sin(\theta_6 + 1.0\pi)\cos(\theta_1)\cos(\theta_7) \\
(\text{a}\dot{}\text{x}) &: (-\sin(\theta_2)\sin(\theta_4) - \cos(\theta_2)\cos(\theta_3)\cos(\theta_4))\sin(\theta_5) - \sin(\theta_3)\cos(\theta_2)\cos(\theta_5) \\
&= -\sin(\theta_7)\cos(\theta_1) \\
(\text{p}\dot{}\text{x}) &: L\,(2.5\,(-\sin(\theta_2)\sin(\theta_4) - \cos(\theta_2)\cos(\theta_3)\cos(\theta_4))\cos(\theta_5) - 2.5\sin(\theta_2)\sin(\theta_4) + 2.5\sin(\theta_3)\sin(\theta_5)\cos(\theta_2) - 2.5\cos(\theta_2)\cos(\theta_3)\cos(\theta_4) + 2.0\cos(\theta_2)) \\
&= L\,(2.5\sin(\theta_7)\cos(\theta_1) - 2.0\cos(\theta_1) - 1.0) \\
(\text{n}\dot{}\text{y}) &: (-\sin(\theta_2)\cos(\theta_3)\cos(\theta_4) + \sin(\theta_4)\cos(\theta_2))\cos(\theta_5) + \sin(\theta_2)\sin(\theta_3)\sin(\theta_5) \\
&= -\sin(\theta_1)\cos(\theta_7)\cos(\theta_6 + 1.0\pi) + \sin(\theta_6 + 1.0\pi)\cos(\theta_1) \\
(\text{o}\dot{}\text{y}) &: \sin(\theta_2)\sin(\theta_4)\cos(\theta_3) + \cos(\theta_2)\cos(\theta_4) \\
&= -\sin(\theta_1)\sin(\theta_6 + 1.0\pi)\cos(\theta_7) - \cos(\theta_1)\cos(\theta_6 + 1.0\pi) \\
(\text{a}\dot{}\text{y}) &: (-\sin(\theta_2)\cos(\theta_3)\cos(\theta_4) + \sin(\theta_4)\cos(\theta_2))\sin(\theta_5) - \sin(\theta_2)\sin(\theta_3)\cos(\theta_5) \\
&= \sin(\theta_1)\sin(\theta_7) \\
(\text{p}\dot{}\text{y}) &: L\,(2.5\,(-\sin(\theta_2)\cos(\theta_3)\cos(\theta_4) + \sin(\theta_4)\cos(\theta_2))\cos(\theta_5) + 2.5\sin(\theta_2)\sin(\theta_3)\sin(\theta_5) - 2.5\sin(\theta_2)\cos(\theta_3)\cos(\theta_4) + 2.0\sin(\theta_2) + 2.5\sin(\theta_4)\cos(\theta_2)) \\
&= L\,(-2.5\sin(\theta_1)\sin(\theta_7) + 2.0\sin(\theta_1)) \\
(\text{n}\dot{}\text{z}) &: \sin(\theta_3)\cos(\theta_4)\cos(\theta_5) + \sin(\theta_5)\cos(\theta_3) \\
&= -\sin(\theta_7)\cos(\theta_6 + 1.0\pi) \\
(\text{o}\dot{}\text{z}) &: -\sin(\theta_3)\sin(\theta_4) \\
&= \sin(\theta_6)\sin(\theta_7) \\
(\text{a}\dot{}\text{z}) &: \sin(\theta_3)\sin(\theta_5)\cos(\theta_4) - \cos(\theta_3)\cos(\theta_5) \\
&= -\cos(\theta_7) \\
(\text{p}\dot{}\text{z}) &: L\,(2.5\sin(\theta_3)\cos(\theta_4)\cos(\theta_5) + 2.5\sin(\theta_3)\cos(\theta_4) + 2.5\sin(\theta_5)\cos(\theta_3) + 5.0) \\
&= 2.5L\cos(\theta_7)
\end{aligned}
$$

3 Permutation 3: $T_3T_4T_5T_6 = T_2^{-1}T_1^{-1}T_7^{-1}$

$$
\begin{aligned}
(\text{n\^{}x}) &: -\left(\sin\left(\theta_3\right)\sin\left(\theta_5\right) - \cos\left(\theta_3\right)\cos\left(\theta_4\right)\cos\left(\theta_5\right)\right)\cos\left(\theta_6\right) - \sin\left(\theta_4\right)\sin\left(\theta_6\right)\cos\left(\theta_3\right) \\
&= \cos\left(\theta_7\right)\cos\left(\theta_1+\theta_2\right) \\
(\text{o\^{}x}) &: \sin\left(\theta_3\right)\cos\left(\theta_5\right) + \sin\left(\theta_5\right)\cos\left(\theta_3\right)\cos\left(\theta_4\right) \\
&= \sin\left(\theta_7\right)\cos\left(\theta_1+\theta_2\right) \\
(\text{a\^{}x}) &: \left(\sin\left(\theta_3\right)\sin\left(\theta_5\right) - \cos\left(\theta_3\right)\cos\left(\theta_4\right)\cos\left(\theta_5\right)\right)\sin\left(\theta_6\right) - \sin\left(\theta_4\right)\cos\left(\theta_3\right)\cos\left(\theta_6\right) \\
&= -\sin\left(\theta_1+\theta_2\right) \\
(\text{p\^{}x}) &: L\left(2.5\sqrt{2}\sin\left(\theta_3+1.0\pi\right)\cos\left(\theta_5+\frac{\pi}{4}\right) + 2.5\sqrt{2}\sin\left(\theta_5+\frac{\pi}{4}\right)\cos\left(\theta_4\right)\cos\left(\theta_3+1.0\pi\right) + 2.5\cos\left(\theta_4\right)\cos\left(\theta_3+1.0\pi\right)\right) \\
&= L\left(-1.0\cos\left(\theta_2\right) - 2.0\cos\left(\theta_1+\theta_2\right) - 2.0\right) \\
(\text{n\^{}y}) &: -\left(-\sin\left(\theta_3\right)\cos\left(\theta_4\right)\cos\left(\theta_5\right) - \sin\left(\theta_5\right)\cos\left(\theta_3\right)\right)\cos\left(\theta_6\right) - \sin\left(\theta_3\right)\sin\left(\theta_4\right)\sin\left(\theta_6\right) \\
&= \sin\left(\theta_7\right) \\
(\text{o\^{}y}) &: \sin\left(\theta_3\right)\sin\left(\theta_5\right)\cos\left(\theta_4\right) - \cos\left(\theta_3\right)\cos\left(\theta_5\right) \\
&= -\cos\left(\theta_7\right) \\
(\text{a\^{}y}) &: \left(-\sin\left(\theta_3\right)\cos\left(\theta_4\right)\cos\left(\theta_5\right) - \sin\left(\theta_5\right)\cos\left(\theta_3\right)\right)\sin\left(\theta_6\right) - \sin\left(\theta_3\right)\sin\left(\theta_4\right)\cos\left(\theta_6\right) \\
&= 0 \\
(\text{p\^{}y}) &: L\left(2.5\sqrt{2}\sin\left(\theta_3+1.0\pi\right)\sin\left(\theta_5+\frac{\pi}{4}\right)\cos\left(\theta_4\right) + 2.5\sin\left(\theta_3+1.0\pi\right)\cos\left(\theta_4\right) - 2.5\sqrt{2}\cos\left(\theta_3+1.0\pi\right)\cos\left(\theta_5+\frac{\pi}{4}\right)\right) \\
&= 5.0L \\
(\text{n\^{}z}) &: -\sin\left(\theta_4\right)\cos\left(\theta_5\right)\cos\left(\theta_6\right) - \sin\left(\theta_6\right)\cos\left(\theta_4\right) \\
&= -\sin\left(\theta_1+\theta_2\right)\cos\left(\theta_7\right) \\
(\text{o\^{}z}) &: -\sin\left(\theta_4\right)\sin\left(\theta_5\right) \\
&= -\sin\left(\theta_7\right)\sin\left(\theta_1+\theta_2\right) \\
(\text{a\^{}z}) &: \sin\left(\theta_4\right)\sin\left(\theta_6\right)\cos\left(\theta_5\right) - \cos\left(\theta_4\right)\cos\left(\theta_6\right) \\
&= -\cos\left(\theta_1+\theta_2\right) \\
(\text{p\^{}z}) &: L\left(2.5\sqrt{2}\sin\left(\theta_4\right)\sin\left(\theta_5+\frac{\pi}{4}\right) + 2.5\sin\left(\theta_4\right)\right) \\
&= L\left(1.0\sin\left(\theta_2\right) + 2.0\sin\left(\theta_1+\theta_2\right)\right)
\end{aligned}
$$

4 Permutation 4: $T_4T_5T_6T_7 = T_3^{-1}T_2^{-1}T_1^{-1}$

$$
\begin{aligned}
(\mathrm{n}\dot{}\mathrm{x}) &: (\sin(\theta_4)\sin(\theta_6) - \cos(\theta_4)\cos(\theta_5)\cos(\theta_6))\cos(\theta_7) - \sin(\theta_5)\sin(\theta_7)\cos(\theta_4) \\
&= \cos(\theta_1+\theta_2)\cos(\theta_3+1.0\pi) \\
(\mathrm{o}\dot{}\mathrm{x}) &: -\sin(\theta_4)\cos(\theta_6) - \sin(\theta_6)\cos(\theta_4)\cos(\theta_5) \\
&= \sin(\theta_1+\theta_2)\cos(\theta_3+1.0\pi) \\
(\mathrm{a}\dot{}\mathrm{x}) &: -(\sin(\theta_4)\sin(\theta_6) - \cos(\theta_4)\cos(\theta_5)\cos(\theta_6))\sin(\theta_7) - \sin(\theta_5)\cos(\theta_4)\cos(\theta_7) \\
&= \sin(\theta_3) \\
(\mathrm{p}\dot{}\mathrm{x}) &: L\left(-2.0\sin(\theta_4)\sin(\theta_6+1.0\pi)\cos(\theta_7) - 2.0\sin(\theta_5)\sin(\theta_7)\cos(\theta_4) + 2.5\sqrt{2}\sin\left(\theta_5+\frac{\pi}{4}\right)\cos(\theta_4) + 2.0\cos(\theta_4)\cos(\theta_5)\cos(\theta_7)\cos(\theta_6+1.0\pi) + 2.5\cos(\theta_4)\right) \\
&= L\left(5.0\sin(\theta_3+1.0\pi) - 1.0\cos(\theta_2)\cos(\theta_3+1.0\pi) - 2.0\cos(\theta_3+1.0\pi)\right) \\
(\mathrm{n}\dot{}\mathrm{y}) &: (-\sin(\theta_4)\cos(\theta_5)\cos(\theta_6) - \sin(\theta_6)\cos(\theta_4))\cos(\theta_7) - \sin(\theta_4)\sin(\theta_5)\sin(\theta_7) \\
&= -\sin(\theta_1+\theta_2) \\
(\mathrm{o}\dot{}\mathrm{y}) &: -\sin(\theta_4)\sin(\theta_6)\cos(\theta_5) + \cos(\theta_4)\cos(\theta_6) \\
&= \cos(\theta_1+\theta_2) \\
(\mathrm{a}\dot{}\mathrm{y}) &: -(-\sin(\theta_4)\cos(\theta_5)\cos(\theta_6) - \sin(\theta_6)\cos(\theta_4))\sin(\theta_7) - \sin(\theta_4)\sin(\theta_5)\cos(\theta_7) \\
&= 0 \\
(\mathrm{p}\dot{}\mathrm{y}) &: L\left(-2.0\sin(\theta_4)\sin(\theta_5)\sin(\theta_7) + 2.5\sqrt{2}\sin(\theta_4)\sin\left(\theta_5+\frac{\pi}{4}\right) + 2.0\sin(\theta_4)\cos(\theta_5)\cos(\theta_7)\cos(\theta_6+1.0\pi) + 2.5\sin(\theta_4) + 2.0\sin(\theta_6+1.0\pi)\cos(\theta_4)\cos(\theta_7)\right) \\
&= 1.0L\sin(\theta_2) \\
(\mathrm{n}\dot{}\mathrm{z}) &: \sin(\theta_5)\cos(\theta_6)\cos(\theta_7) - \sin(\theta_7)\cos(\theta_5) \\
&= \sin(\theta_3+1.0\pi)\cos(\theta_1+\theta_2) \\
(\mathrm{o}\dot{}\mathrm{z}) &: \sin(\theta_5)\sin(\theta_6) \\
&= \sin(\theta_1+\theta_2)\sin(\theta_3+1.0\pi) \\
(\mathrm{a}\dot{}\mathrm{z}) &: -\sin(\theta_5)\sin(\theta_7)\cos(\theta_6) - \cos(\theta_5)\cos(\theta_7) \\
&= -\cos(\theta_3) \\
(\mathrm{p}\dot{}\mathrm{z}) &: L\left(-2.0\sin(\theta_5)\cos(\theta_7)\cos(\theta_6+1.0\pi) - 2.0\sin(\theta_7)\cos(\theta_5) + 2.5\sqrt{2}\cos\left(\theta_5+\frac{\pi}{4}\right)\right) \\
&= L\left(-1.0\sin(\theta_3+1.0\pi)\cos(\theta_2) - 2.0\sin(\theta_3+1.0\pi) - 5.0\cos(\theta_3+1.0\pi)\right)
\end{aligned}
$$

5 Permutation 5: $T_5T_6T_7T_1 = T_4^{-1}T_3^{-1}T_2^{-1}$

$$
\begin{aligned}
(\text{n}\dot{}\text{x}) &: (-\sin(\theta_5)\sin(\theta_7) - \cos(\theta_5)\cos(\theta_6)\cos(\theta_7))\cos(\theta_1) - \sin(\theta_1)\sin(\theta_6)\cos(\theta_5) \\
&= -\sin(\theta_2)\sin(\theta_4) + \cos(\theta_2)\cos(\theta_4)\cos(\theta_3 + 1.0\pi) \\
(\text{o}\dot{}\text{x}) &: -(-\sin(\theta_5)\sin(\theta_7) - \cos(\theta_5)\cos(\theta_6)\cos(\theta_7))\sin(\theta_1) - \sin(\theta_6)\cos(\theta_1)\cos(\theta_5) \\
&= \sin(\theta_2)\cos(\theta_4)\cos(\theta_3 + 1.0\pi) + \sin(\theta_4)\cos(\theta_2) \\
(\text{a}\dot{}\text{x}) &: -\sin(\theta_5)\cos(\theta_7) + \sin(\theta_7)\cos(\theta_5)\cos(\theta_6) \\
&= -\sin(\theta_3 + 1.0\pi)\cos(\theta_4) \\
(\text{p}\dot{}\text{x}) &: L\,(1.0\,(-\sin(\theta_5)\sin(\theta_7) - \cos(\theta_5)\cos(\theta_6)\cos(\theta_7))\cos(\theta_1) - 1.0\sin(\theta_1)\sin(\theta_6)\cos(\theta_5) - 2.0\sin(\theta_5)\sin(\theta_7) + 2.5\sin(\theta_5) - 2.0\cos(\theta_5)\cos(\theta_6)\cos(\theta_7) + 2.5\cos(\theta_5)) \\
&= L\,(5.0\sin(\theta_3 + 1.0\pi)\cos(\theta_4) - 2.0\cos(\theta_4)\cos(\theta_3 + 1.0\pi) - 2.5) \\
(\text{n}\dot{}\text{y}) &: (-\sin(\theta_5)\cos(\theta_6)\cos(\theta_7) + \sin(\theta_7)\cos(\theta_5))\cos(\theta_1) - \sin(\theta_1)\sin(\theta_5)\sin(\theta_6) \\
&= -\sin(\theta_3 + 1.0\pi)\cos(\theta_2) \\
(\text{o}\dot{}\text{y}) &: -(-\sin(\theta_5)\cos(\theta_6)\cos(\theta_7) + \sin(\theta_7)\cos(\theta_5))\sin(\theta_1) - \sin(\theta_5)\sin(\theta_6)\cos(\theta_1) \\
&= \sin(\theta_2)\sin(\theta_3) \\
(\text{a}\dot{}\text{y}) &: \sin(\theta_5)\sin(\theta_7)\cos(\theta_6) + \cos(\theta_5)\cos(\theta_7) \\
&= \cos(\theta_3) \\
(\text{p}\dot{}\text{y}) &: L\,(1.0\,(-\sin(\theta_5)\cos(\theta_6)\cos(\theta_7) + \sin(\theta_7)\cos(\theta_5))\cos(\theta_1) - 1.0\sin(\theta_1)\sin(\theta_5)\sin(\theta_6) - 2.0\sin(\theta_5)\cos(\theta_6)\cos(\theta_7) + 2.5\sin(\theta_5) + 2.0\sin(\theta_7)\cos(\theta_5) - 2.5\cos(\theta_5)) \\
&= L\,(-2.0\sin(\theta_3) - 5.0\cos(\theta_3)) \\
(\text{n}\dot{}\text{z}) &: \sin(\theta_1)\cos(\theta_6) - \sin(\theta_6)\cos(\theta_1)\cos(\theta_7) \\
&= -\sin(\theta_2)\cos(\theta_4) - \sin(\theta_4)\cos(\theta_2)\cos(\theta_3 + 1.0\pi) \\
(\text{o}\dot{}\text{z}) &: \sin(\theta_1)\sin(\theta_6)\cos(\theta_7) + \cos(\theta_1)\cos(\theta_6) \\
&= -\sin(\theta_2)\sin(\theta_4)\cos(\theta_3 + 1.0\pi) + \cos(\theta_2)\cos(\theta_4) \\
(\text{a}\dot{}\text{z}) &: \sin(\theta_6)\sin(\theta_7) \\
&= -\sin(\theta_3)\sin(\theta_4) \\
(\text{p}\dot{}\text{z}) &: L\,(1.0\sin(\theta_1)\cos(\theta_6) - 1.0\sin(\theta_6)\cos(\theta_1)\cos(\theta_7) - 2.0\sin(\theta_6)\cos(\theta_7)) \\
&= L\,(-5.0\sin(\theta_4)\sin(\theta_3 + 1.0\pi) + 2.0\sin(\theta_4)\cos(\theta_3 + 1.0\pi))
\end{aligned}
$$

6 Permutation 6: $T_6T_7T_1T_2 = T_5^{-1}T_4^{-1}T_3^{-1}$

$$
\begin{aligned}
(\text{n}\dot{}\text{x}) &: \sin(\theta_1+\theta_2)\sin(\theta_6+1.0\pi)+\cos(\theta_7)\cos(\theta_1+\theta_2)\cos(\theta_6+1.0\pi)\\
&= -\sin(\theta_5)\sin(\theta_3+1.0\pi)+\cos(\theta_4)\cos(\theta_5)\cos(\theta_3+1.0\pi)\\
(\text{o}\dot{}\text{x}) &: -\sin(\theta_7)\cos(\theta_6)\\
&= \sin(\theta_5)\cos(\theta_3+1.0\pi)+\sin(\theta_3+1.0\pi)\cos(\theta_4)\cos(\theta_5)\\
(\text{a}\dot{}\text{x}) &: -\sin(\theta_1+\theta_2)\cos(\theta_7)\cos(\theta_6+1.0\pi)+\sin(\theta_6+1.0\pi)\cos(\theta_1+\theta_2)\\
&= \sin(\theta_4)\cos(\theta_5)\\
(\text{p}\dot{}\text{x}) &: L\,(1.0\sin(\theta_1)\sin(\theta_6+1.0\pi)-5.0\sin(\theta_7)\cos(\theta_6+1.0\pi)+2.0\sin(\theta_1+\theta_2)\sin(\theta_6+1.0\pi)+1.0\cos(\theta_1)\cos(\theta_7)\cos(\theta_6+1.0\pi)+2.0\cos(\theta_7)\cos(\theta_1+\theta_2)\cos(\theta_6+1.0\pi)+2.0\cos(\theta_7)\cos(\theta_6+1.0\pi))\\
&= L\,(-2.5\cos(\theta_5)-2.5)\\
(\text{n}\dot{}\text{y}) &: -\sin(\theta_1+\theta_2)\cos(\theta_6+1.0\pi)+\sin(\theta_6+1.0\pi)\cos(\theta_7)\cos(\theta_1+\theta_2)\\
&= -\sin(\theta_4)\cos(\theta_3+1.0\pi)\\
(\text{o}\dot{}\text{y}) &: -\sin(\theta_6)\sin(\theta_7)\\
&= \sin(\theta_3)\sin(\theta_4)\\
(\text{a}\dot{}\text{y}) &: -\sin(\theta_1+\theta_2)\sin(\theta_6+1.0\pi)\cos(\theta_7)-\cos(\theta_1+\theta_2)\cos(\theta_6+1.0\pi)\\
&= \cos(\theta_4)\\
(\text{p}\dot{}\text{y}) &: L\,(-1.0\sin(\theta_1)\cos(\theta_6+1.0\pi)-5.0\sin(\theta_7)\sin(\theta_6+1.0\pi)-2.0\sin(\theta_1+\theta_2)\cos(\theta_6+1.0\pi)+1.0\sin(\theta_6+1.0\pi)\cos(\theta_1)\cos(\theta_7)+2.0\sin(\theta_6+1.0\pi)\cos(\theta_7)\cos(\theta_1+\theta_2)+2.0\sin(\theta_6+1.0\pi)\cos(\theta_7))\\
&= 0\\
(\text{n}\dot{}\text{z}) &: -\sin(\theta_7)\cos(\theta_1+\theta_2)\\
&= \sin(\theta_5)\cos(\theta_4)\cos(\theta_3+1.0\pi)+\sin(\theta_3+1.0\pi)\cos(\theta_5)\\
(\text{o}\dot{}\text{z}) &: \cos(\theta_7)\\
&= \sin(\theta_5)\sin(\theta_3+1.0\pi)\cos(\theta_4)-\cos(\theta_5)\cos(\theta_3+1.0\pi)\\
(\text{a}\dot{}\text{z}) &: \sin(\theta_7)\sin(\theta_1+\theta_2)\\
&= \sin(\theta_4)\sin(\theta_5)\\
(\text{p}\dot{}\text{z}) &: L\,(-1.0\sin(\theta_7)\cos(\theta_1)-2.0\sin(\theta_7)\cos(\theta_1+\theta_2)-2.0\sin(\theta_7)-5.0\cos(\theta_7)+2.5)\\
&= -2.5L\sin(\theta_5)
\end{aligned}
$$

7 Permutation 7: $T_7T_1T_2T_3 = T_6^{-1}T_5^{-1}T_4^{-1}$

$$
\begin{aligned}
(\text{n}\dot{}\text{x}) &: \sin(\theta_7)\sin(\theta_3 + 1.0\pi) + \cos(\theta_7)\cos(\theta_1 + \theta_2)\cos(\theta_3 + 1.0\pi) \\
&= -\sin(\theta_4)\sin(\theta_6 + 1.0\pi) + \cos(\theta_4)\cos(\theta_5)\cos(\theta_6 + 1.0\pi) \\
(\text{o}\dot{}\text{x}) &: -\sin(\theta_1 + \theta_2)\cos(\theta_7) \\
&= \sin(\theta_4)\cos(\theta_5)\cos(\theta_6 + 1.0\pi) + \sin(\theta_6 + 1.0\pi)\cos(\theta_4) \\
(\text{a}\dot{}\text{x}) &: -\sin(\theta_7)\cos(\theta_3 + 1.0\pi) + \sin(\theta_3 + 1.0\pi)\cos(\theta_7)\cos(\theta_1 + \theta_2) \\
&= -\sin(\theta_5)\cos(\theta_6 + 1.0\pi) \\
(\text{p}\dot{}\text{x}) &: L(-5.0\sin(\theta_7) + 1.0\cos(\theta_1)\cos(\theta_7) + 2.0\cos(\theta_7)\cos(\theta_1 + \theta_2) + 2.0\cos(\theta_7)) \\
&= L(-2.5\cos(\theta_5)\cos(\theta_6 + 1.0\pi) - 2.5\cos(\theta_6 + 1.0\pi)) \\
(\text{n}\dot{}\text{y}) &: \sin(\theta_7)\cos(\theta_1 + \theta_2)\cos(\theta_3 + 1.0\pi) - \sin(\theta_3 + 1.0\pi)\cos(\theta_7) \\
&= -\sin(\theta_5)\cos(\theta_4) \\
(\text{o}\dot{}\text{y}) &: -\sin(\theta_7)\sin(\theta_1 + \theta_2) \\
&= -\sin(\theta_4)\sin(\theta_5) \\
(\text{a}\dot{}\text{y}) &: \sin(\theta_7)\sin(\theta_3 + 1.0\pi)\cos(\theta_1 + \theta_2) + \cos(\theta_7)\cos(\theta_3 + 1.0\pi) \\
&= -\cos(\theta_5) \\
(\text{p}\dot{}\text{y}) &: L(1.0\sin(\theta_7)\cos(\theta_1) + 2.0\sin(\theta_7)\cos(\theta_1 + \theta_2) + 2.0\sin(\theta_7) + 5.0\cos(\theta_7)) \\
&= L(2.5\sin(\theta_5) + 2.5) \\
(\text{n}\dot{}\text{z}) &: -\sin(\theta_1 + \theta_2)\cos(\theta_3 + 1.0\pi) \\
&= -\sin(\theta_4)\cos(\theta_6 + 1.0\pi) - \sin(\theta_6 + 1.0\pi)\cos(\theta_4)\cos(\theta_5) \\
(\text{o}\dot{}\text{z}) &: -\cos(\theta_1 + \theta_2) \\
&= -\sin(\theta_4)\sin(\theta_6 + 1.0\pi)\cos(\theta_5) + \cos(\theta_4)\cos(\theta_6 + 1.0\pi) \\
(\text{a}\dot{}\text{z}) &: -\sin(\theta_1 + \theta_2)\sin(\theta_3 + 1.0\pi) \\
&= -\sin(\theta_5)\sin(\theta_6) \\
(\text{p}\dot{}\text{z}) &: L(-1.0\sin(\theta_1) - 2.0\sin(\theta_1 + \theta_2)) \\
&= L(2.5\sin(\theta_6 + 1.0\pi)\cos(\theta_5) + 2.5\sin(\theta_6 + 1.0\pi))
\end{aligned}
$$